A one-body Toda–Boussinesq dispersion test: the bad-equation continuation, its resonance, and a numerical mismatch

Research note for Gustav Delius

4 November 2026

Abstract

(c) With the inherited classical units and exact dressed momentum, the saved Toda data give a fitted small-amplitude one-loop coefficient $F_T/|P_B| = 0.15912 \pm 0.00334$. The uncertainty is a deliberately broad fit-sensitivity envelope, not a rigorous error bound. Direct group-velocity data give 0.15906 ± 0.00997. The good-equation benchmark is $-2/\pi$. A dimensionally consistent *formal continuation* gives instead $+2/\pi = 0.63661977$ for the bad equation: a discrepancy 0.47750 ± 0.00334, approximately a factor of four.

(c) The real stable-band trace of the bad equation is different again: it adds $-\frac{\sqrt{3}}{2}(\sinh^2\beta + \frac{3}{4})$ to the formally continued answer. Its normal-ordering energy has sign $-3\sinh\beta/\pi$, correcting the opposite sign in the older bad-equation note. A differential combination removes constant energy, momentum-origin and mass-counterterm ambiguities and retains the same cubic coefficient $+2/\pi$.

(d) There is no bad-equation real-frequency trace over the full continuum with just the good-equation assumptions. Neither continuation nor deletion of unstable modes follows from those assumptions. Thus the results disfavor the specified conditional targets within the tested numerical extrapolations; they do not refute a completed quantum continuum theory or distinguish the mode and canonical charge pairs. The physical charge identification remains inconclusive because the determinant/domain and quantum continuum matching are not established, and the required interchange of limits is unproved.

1 What is being tested, and what is conditional

Labels are (a) literature proof, (b) literature heuristic, (c) project derivation/check, and (d) unestablished identification. All new mathematical formulas below are checked by `onebody_continuum.py` or `onebody_toda.py`, as indicated by section. Exact algebra and numerical evidence are distinguished. The scripts use SymPy and 65-digit mpmath; high precision in the data analysis does not increase the precision of the inherited double-precision sea solves.

The inputs read are the complete internal notes `det_projected_oneloop.tex`, `frame_dictionary.tex`, `toda_route.tex`, and `oneloop.tex`, together with the relevant checking implementations and saved `frame_*.json` data. No external full text or abstract was read for this note; no external reference, DOI, arXiv identifier, or literature theorem is asserted. In particular the old note's identification of a stable-band quantum domain with an inverse-scattering reduction is not adopted as a theorem.

Loop order and fixed momentum. (c) (`onebody_continuum.py`, `onebody_toda.py`). We write

$$E(P) = E_0(P) + \hbar F(P) + O(\hbar^2), \qquad F = e - vp = -L_1, \qquad E_0' = vP_0'. \tag{1}$$

Thus F is the order-$\hbar^0$ contribution to $E/\hbar$, and the order-$\hbar$ contribution to energy. A tangent relabeling $(e, p) \mapsto (e + v\delta p, p + \delta p)$ leaves F invariant. For the canonical pair $p_{\rm can} = \partial_v L_1$, $e_{\rm can} = vp_{\rm can} - L_1$, the same F follows. Consequently a one-body fixed-momentum experiment cannot decide which of these two pairs represents the renormalized charges.

Exact qualifications inherited from the det package. The good benchmark uses its specified vacuum Wick ordering, no added finite multiple of M, and an equal-mode-count trace *including* the cutoff endpoint. Dropping that endpoint leaves an uncanceled linear divergence. **(d)** The identification of this trace with the absolutely normalized projected determinant is not proved. The instantaneous symplectic projector does not preserve fixed potential endpoints; shift and width tangents have divergent kinetic norms. A common boundary/constraint evolution problem, its Schur complement, nonzero-frequency completeness, and time-endpoint determinant phase remain missing. Differentiating the on-shell L_1 changes the endpoint sector and does not prove the canonical charge identification. The good real-energy benchmark is restricted to the light branch $0 < \alpha < \pi/3$. Nothing in this note resolves these issues.

No new physical prescription is selected. The older `oneloop.tex` records a PI choice of stable-band reduction and an additional momentum-endpoint prescription. The present request asks for the same method as the newer good-equation package. These are distinct inputs. Below we evaluate both the formal continuation and a fully stated stable-band diagnostic; neither is silently selected as the physical bad determinant. An additional prescription would require the PI's choice.

2 Bad versus good: the dictionary and the sign

Classical dictionary. (c) (`onebody_continuum.py`). For the bad equation the stipulated symplectic normalization gives

$$\mathcal{L}_b = \tfrac{1}{2}\phi_t^2 - \tfrac{1}{2}\phi_x^2 - \phi_x^3 + \tfrac{1}{2}\phi_{xx}^2, \qquad u_0 = \tfrac{k^2}{2}\operatorname{sech}^2(k\xi/2), \tag{2}$$

$$k = \sinh\beta, \qquad v = \epsilon\cosh\beta, \qquad \ell_b = \epsilon\cosh 3\beta, \tag{3}$$

$$M = 2k, \qquad \Pi = -2vk, \qquad P_0 = \tfrac{2}{3}vk^3, \tag{4}$$

$$E_0 = \tfrac{2}{3}k^3 + \tfrac{8}{15}k^5, \qquad L_0 = vP_0 - E_0 = \tfrac{2}{15}k^5. \tag{5}$$

The good values are $k_g = \sin\alpha$, $v_g = \epsilon\cos\alpha$, $\ell_g = -\epsilon\cos 3\alpha$, $E_{g,0} = 2k_g^3/3 - 8k_g^5/15$. Formally $\alpha = i\beta$ gives $k_g = ik$, $v_g = v$, and $\ell_g = -\ell_b$. The sign of the spectral parameter is part of the dictionary; it cannot be omitted.

Why substituting $\alpha = i\beta$ alone is insufficient. (c) (`onebody_continuum.py`). Introduce the algebraic family

$$\mathcal{L}_\sigma = \tfrac{1}{2}\phi_t^2 - \tfrac{1}{2}\phi_x^2 + \sigma\phi_x^3 - \tfrac{\sigma}{2}\phi_{xx}^2, \qquad \sigma = c^2. \tag{6}$$

For positive c, put $(x,t)_\sigma = c(x,t)_g$ and $\phi_\sigma = \phi_g/c$. Direct substitution gives

$$S_\sigma = c^{-2}S_g, \qquad \hbar_g = c^2\hbar_\sigma, \qquad (E_0,P_0)_\sigma = c^{-3}(E_0,P_0)_g, \qquad F_\sigma = c^{-1}F_g. \tag{7}$$

Continue these algebraic relations to $c = i$, using branches continuous from the light good soliton. In particular $\hbar_g = -\hbar_b$ and $u_g = -u_b$ in this *formal* calculation. This is not a real negative Planck constant or a unitary equivalence of quantum theories. **(d)** Continuing the renormalized trace without extra contour contributions is an additional assumption, not a consequence of the classical dictionary.

The formal continued target. (c) Conditional on that assumption (`onebody_continuum.py`). The analytic good fixed-P expression on its light branch is

$$F_g = \frac{\sqrt{3}}{\pi}\left(\frac{3}{4} - k_g^2\right)\operatorname{artanh}\frac{2k_g}{\sqrt{3}} - \frac{3k_g}{2\pi}. \tag{8}$$

With (7) its bad analogue is

$$\boxed{F_{\rm ac}(\beta) = -iF_g(i\beta) = \frac{\sqrt{3}}{\pi}\left(k^2 + \frac{3}{4}\right)\arctan\frac{2k}{\sqrt{3}} - \frac{3k}{2\pi}.} \tag{9}$$

It obeys $L_{1,\mathrm{ac}} = -F_{\mathrm{ac}}$ and

$$p_{\mathrm{can,ac}} = -\frac{2\epsilon\sqrt{3}}{\pi}\cosh\beta \arctan\frac{2\sinh\beta}{\sqrt{3}}, \tag{10}$$

$$F_{\mathrm{ac}} = \frac{4}{3\pi}\beta^3 + \frac{14}{45\pi}\beta^5 + O(\beta^7), \qquad \frac{F_{\mathrm{ac}}}{|P_0|} \longrightarrow +\frac{2}{\pi}. \tag{11}$$

The good answer instead has $F_g = -4\alpha^3/(3\pi) + 14\alpha^5/(45\pi) + O(\alpha^7)$ and ratio $-2/\pi$. Thus even this specific continuation does *not* give the same small-amplitude coefficient. Forgetting the action/Planck scaling would reverse the continued sign. Shared long-wave classical behavior does not by itself identify a trace integrating all wavelengths.

3 The real bad-equation trace and its extra resonance

All calculations in this section are **(c)** (`onebody_continuum.py`); interpreting their trace as a projected quantum energy remains **(d)** .

The full-line obstruction. Linear vacuum modes have $\omega^2 = q^2 - q^4$. For $|q| > 1$ there is no real positive-frequency oscillator vacuum of the good-package kind. For example the continued Hirota transmission has modulus 1.8388478863 at $\beta = 0.7, q = 1.3, \epsilon = +1$. A real scattering phase over the entire line is therefore unavailable. There is no unique bad analogue *with only* the same conditional ingredients as the good package.

Explicit stable-band diagnostic. Restrict to $|q| < 1$ with $\omega = |q|\sqrt{1-q^2}$. The Hirota polynomial and transmission are

$$\mathcal{H}(K, W) = W^2 - K^2 - K^4, \tag{12}$$

$$A(q) = -\frac{\mathcal{H}(k - iq, kv - i\omega)}{\mathcal{H}(k + iq, kv + i\omega)} = \frac{Z + iX}{Z - iX}, \tag{13}$$

$$X = 3kq, \qquad Z = v\ \mathrm{sgn}(q)\sqrt{1-q^2} - 1 - 2k^2 + 2q^2. \tag{14}$$

The potential mode $f = \partial_\xi[e^{iq\xi}(1 + Ae^{k\xi})/(1 + e^{k\xi})]$ satisfies the bad potential pencil

$$[-\partial_\xi^4 + (v^2 - 1)\partial_\xi^2 - 6\partial_\xi(u_0\partial_\xi) + 2iv\nu\partial_\xi - \nu^2]f = 0, \qquad \nu = \omega - vq. \tag{15}$$

The script independently checks this residual, $|A| = 1$ in the stable band, and the differentiated phase.

Let $q = \sin t$, $-\pi/2 < t < \pi/2$, and choose the continuous logarithm on each half-band with $\theta(0^{\pm}) = 0$ at fixed $\beta > 0$. Then

$$\theta_t = \frac{2(ZX_t - XZ_t)}{Z^2 + X^2}. \tag{16}$$

Retaining the equal-mode-count endpoint in the co-moving trace would give

$$F_{\mathrm{band}} = \frac{1}{4\pi}\int_{-1}^{1}(\omega - vq)\theta'(q)\,\mathrm{d}q - \frac{[(\omega - vq)\theta]_{-1}^{1}}{4\pi} - \frac{3k}{\pi}. \tag{17}$$

For the stated continuous branches $\theta(1) + \theta(-1) = 0$ and the displayed endpoint vanishes. This is an explicitly checked branch diagnostic, not a derivation of finite-box endpoint quantization. Changing half-band branches can change that endpoint and the momentum origin.

Normal-ordering sign. Here $\langle\phi_x^2\rangle/\hbar = 1/(2\pi)$ on the stable band, but the bad Hamiltonian contains $+u^3$, whereas the good one contains $-u^3$. Hence $:u^3: = u^3 - 3\langle u^2\rangle u$ gives

$$F_{\mathrm{ct,b}} = -\frac{3M}{2\pi} = -\frac{3k}{\pi}. \tag{18}$$

The older bad note's $+3k/\pi$ is the good Hamiltonian's sign. Using that old sign would add $6k/\pi$ to (17); it is a mass-counterterm difference, not the stipulated bad vacuum Wick ordering.

Evaluation, including the short-wave term. Adding the two half-bands and setting $z = \cos t$ reduces the integral before its counterterm to

$$I = \frac{3k}{2\pi} \int_0^1 R_k(z)\, \mathrm{d}z, \tag{19}$$

$$R_k(z) = \frac{2(-4k^2 + 8z^4 - 10z^2 - 1)}{16k^4 + 16k^2z^2 + 8k^2 + 16z^4 - 8z^2 + 1}. \tag{20}$$

An antiderivative vanishing at zero is

$$z - \frac{\sqrt{3}(4k^2+3)}{12k} \left[\arctan \frac{z}{\sqrt{3}k} + \arctan \frac{\sqrt{3}z(4z^2 + 8k^2 - 1)}{3k(4k^2+1)} \right]. \tag{21}$$

At $z = 1$ the sum of angles equals $\pi - 2\arctan(2k/\sqrt{3})$; both sides lie in $(0, \pi)$ and have the same tangent. Thus

$$\boxed{F_{\mathrm{band}} = F_{\mathrm{ac}} - \frac{\sqrt{3}}{2}\left(k^2 + \frac{3}{4}\right).} \tag{22}$$

The independent 65-digit quadratures at $\beta = 0.01, 0.05, 0.2, 0.7$, for both orientations, agree with this expression within 2.4×10^{-66}; exact symbolic differentiation checks the antiderivative. This is a trace identity, not a completeness theorem.

The small-amplitude expansion is

$$F_{\mathrm{band}} = -\frac{3\sqrt{3}}{8} - \frac{\sqrt{3}}{2}\beta^2 + \frac{4}{3\pi}\beta^3 - \frac{\sqrt{3}}{6}\beta^4 + \frac{14}{45\pi}\beta^5 + O(\beta^6). \tag{23}$$

In particular $F_{\mathrm{band}}(0.01) = -0.6496052338424$ and there is no finite $F_{\mathrm{band}}/|P_0|$ limit. The constant $-3\sqrt{3}/8 = -0.6495190528$ is consistent with the short-wave phase winding: at $q_* = -\sqrt{3}/2$ for a right mover, $\omega(q_*) = \sqrt{3}/4$, $\omega'(q_*) = 1$, and $-\frac{1}{2}(\omega(q_*) - q_*) = -3\sqrt{3}/8$. The algebraic term in (22) makes precise why continuing a closed good trace and integrating the real bad stable band are different operations. It cannot be omitted as if analytic continuation automatically preserved the physical mode contour.

4 Toda extraction at fixed dressed momentum

This section is **(c)** (`onebody_toda.py`), numerical where fitted. It uses no two-body phase and no Phase 2i Wiener–Hopf formula.

Units and the exact one-body identities. The inherited dictionary, with $h = \hbar_T$, is

$$\kappa = \sqrt{3}\beta, \quad h = 3b\hbar_B, \quad b = e^{-a/2}, \quad P_B = \frac{a\overline{P}}{6\sqrt{3}b}, \quad E_B = \frac{\overline{E}}{6\sqrt{3}b^2}. \tag{24}$$

It is a leading classical continuum map. In the saved data $a = 1$. The Fredholm identities are inherited checked inputs from `frame_identities.py`; `onebody_toda.py` checks their unit and chain-rule consequences and reanalyzes their saved measurements. The frame package supplies $D = P'(p)$, $V = E'(p)$ and the edge-subtracted $(\overline{P}, \overline{E})$, with zero at $p = q$. Its exact momentum inversion is

$$\overline{P}(q + hu(y)) = P_{T,0}(\kappa) = \frac{4b}{a}(\kappa\cosh\kappa - \sinh\kappa), \qquad \kappa^2 = yh/b, \tag{25}$$

$$E_{T,0}(\kappa) = 2b^2(\sinh\kappa\cosh\kappa - \kappa), \qquad \frac{\mathrm{d}u}{\mathrm{d}y} = \frac{2\sinh\kappa}{aD}, \tag{26}$$

$$\frac{\mathrm{d}\overline{E}}{\mathrm{d}\overline{P}} = \frac{V}{D}, \qquad \frac{\mathrm{d}E_{T,0}}{\mathrm{d}P_{T,0}} = ab\frac{\sinh\kappa}{\kappa}. \tag{27}$$

The finite-amplitude classical Toda dispersion is subtracted *before* any continuum extrapolation. The dictionary reproduces the leading cubic momentum and energy and the fifth-order excess energy above sound; it is not an exact finite-β identification.

Define, exactly as reconstructed from the stored data,

$$C_{E,h}(y) = \frac{3}{a}\frac{\overline{E} - E_{T,0}}{hP_{T,0}}, \tag{28}$$

$$C_{v,h}(y) = \frac{3}{a}\frac{V/D - ab\sinh\kappa/\kappa}{h}. \tag{29}$$

Then $E_B - E_{B,0} = \hbar_B P_B C_{E,h}$ and the quantum velocity correction in the fixed inherited units is $\hbar_B C_{v,h}$. The factor $3/a$ follows from *all three* scales of energy, momentum and Planck's constant. It is not 1, $3b$, or $1/(3b)$. For a smooth edge limit, the chain rule also gives $C_v = C_E + (2y/3)C_E'$.

y	$C_E(y)$	edge spread	$C_v(y)$	edge spread
4	0.160932810	$2.15\,10^{-6}$	0.159681133	$5.68\,10^{-6}$
8	0.160019933	$7.36\,10^{-6}$	0.159427446	$2.03\,10^{-5}$
16	0.159580964	$2.63\,10^{-5}$	0.159289044	$7.33\,10^{-5}$
32	0.159360609	$9.50\,10^{-5}$	0.159193402	$2.59\,10^{-4}$
64	0.159216903	$3.35\,10^{-4}$	0.158986499	$9.68\,10^{-4}$

These are new independent fits of the saved finite-sea measurements, not new sea solves. The script checks the input SHA-256 against the frame analysis and reproduces its energy extrapolations to 10^{-12}.

Error budget and large-y extraction. The raw h values are $.08, .04, .02, .01, .005$, with the frame package's two refined meshes at $.01, .005$. Central edge values use a quadratic in the last three h values. The spread compares the last-two linear, last-four cubic, and last-four $(1, h, h^2, h\log h)$ fits. Measured mesh differences are multiplied by the largest edge-fit weight norm, 15. We then fit $y \geq 4, 8, 16$ through 64, with each of four models:

$$C + A/y, \qquad C + A/y + B/y^2, \qquad C + A/\sqrt{y} + B/y, \qquad C + A/y + B\log(y)/y. \tag{30}$$

All coefficients are free; $1/(2\pi)$ is never supplied to a fit. The central model is $C + A/y$ on $4 \leq y \leq 64$. For each fit, edge and mesh diagnostics are propagated with absolute intercept weights. The final envelope is the maximum over all twelve fits of the displacement from the central intercept plus that fit's propagated edge and mesh diagnostics.

Diagnostic	Energy coefficient	Velocity coefficient
Central intercept	0.159119876	0.159062733
Central propagated edge spread	0.000184053	0.000525545
Central propagated mesh difference	$3.15\,10^{-9}$	$4.71\,10^{-9}$
Largest intercept model/window displacement	0.000208568	0.001051254
Full combined envelope	0.003339590	0.009973693

The wider envelope retains the poorly conditioned three-point $(1, y^{-1/2}, y^{-1})$ extrapolation; it is not discarded to sharpen the result. The fitted coefficient is consistent with $1/(2\pi) = 0.1591549431$, but an exact identification with that number is **(d)** .

Order of limits: the unbounded systematic. At fixed y, $\hbar_B/\beta^2 = 1/y$, so a finite-y table is a quantum crossover, not a one-loop coefficient. The desired semiclassical coefficient takes $h \to 0$ at fixed nonzero momentum and then $\beta \to 0$. The data instead permit $h \to 0$ at fixed y and a subsequent fitted $y \to \infty$. A simultaneous limit must keep $hy/b \to 0$. **(d)** Exchanging these operations requires uniform asymptotic control that the finite data do not supply. No numerical

bound for this systematic, or for untested large-y behaviors, is included in the displayed envelope. It would be incorrect to interpret the discrepancy as a many-sigma proof.

(c) As a separate check of fixed, unscaled momentum, the saved data give the Toda-unit $F_T = 0.007775546, 0.020319171, 0.030098075$ at $\overline{P} = 0.15, 0.4, 0.6$, respectively. These finite-amplitude values reproduce the frame package, but are not sufficiently close to zero to prove the small-amplitude coefficient by themselves.

5 Comparison and convention tests

Numerical comparison. (c) (`onebody_continuum.py`, `onebody_toda.py`). Within the fixed units and the tested fit families,

$$C_{\rm ac} = +2/\pi = 0.6366197724, \tag{31}$$

$$C_T = 0.1591198755 \pm 0.0033395904, \tag{32}$$

$$C_{\rm ac} - C_T = 0.4774998968 \pm 0.0033395904. \tag{33}$$

The nominal ratio is 4.00088. Reusing the good coefficient without a dictionary would instead give a difference -0.7957396479; it does not repair the comparison. In the specified stable-band trace the ratio $F/|P|$ does not have a finite limit at all, by (23).

Which constants matter? (c) (`onebody_continuum.py`). A vacuum-energy subtraction or a chemical-potential offset per edge particle is independent of its momentum and drops out of $\overline{E} = E(p) - E(q)$. It cannot supply a term proportional to $|P|$. An order-$\hbar$ change of energy and momentum origins, and a finite mass counterterm, act on the coefficient on a right-moving branch as

$$F(\beta) \longmapsto F(\beta) + A - B\cosh\beta + 2C\sinh\beta. \tag{34}$$

Thus $\mathrm{d}F/\mathrm{d}P$ removes A, but does *not* by itself remove a momentum-origin change B. Our quoted velocity comparison fixes that origin to the same Fermi edge as the energy. Replacing $P(q)$ by 0 only in the momentum label would be an inconsistent reference change.

A stronger origin- and mass-counterterm-independent diagnostic is

$$\mathcal{I}[F] = \lim_{\beta\to 0^+} \frac{(\partial_\beta^3 - \partial_\beta)F}{(\partial_\beta^3 - \partial_\beta)P_0}. \tag{35}$$

The differential operator annihilates 1, $\cosh\beta$ and $\sinh\beta$ exactly, while its denominator tends to 4. Consequently

$$\mathcal{I}[F_{\rm ac}] = \mathcal{I}[F_{\rm band}] = 2/\pi. \tag{36}$$

For a differentiable asymptotic dispersion $F_T = C_T P_0 + o(\beta^3)$ with the corresponding differentiated remainder small, $\mathcal{I}[F_T] = C_T$. **(d)** That differentiable expansion is a conditional interpretation of the Toda fits, not a measurement of a third derivative at zero. The independent finite-y velocities provide a less derivative-sensitive check in the fixed edge convention.

(c) The stable-band resonance difference is $-\frac{\sqrt{3}}{2}(v^2 - \frac{1}{4})$. Its second v derivative is $-\sqrt{3}$, so it is not an affine energy/momentum-origin shift. Constants can remove its zeroth-order limit, and a momentum-origin change can cancel its leading β^2 piece, but (35) still retains $+2/\pi$. Hence none of the changes in (34) can change the cubic discrepancy. Arbitrary chemical potentials coupled to other nonlinear soliton charges would change the Hamiltonian being compared; they are additional matching data, not merely the edge energy zero.

Quantum sound speed and time normalization. (c) /(d) (`onebody_toda.py`). The saved Toda sound-speed shift in Boussinesq units is

$$s_{1,T} = 0.2387324171, \qquad \text{edge spread } 1.69 \times 10^{-8}, \quad \text{propagated mesh } 5.33 \times 10^{-10}. \tag{37}$$

Subtracting the quantum sound line consistently on the Toda side gives

$$C_T - s_{1,T} = -0.0796125416 \pm 0.00333961, \tag{38}$$

not $\pm 2/\pi$. A change of time normalization or of the background Hamiltonian can add a term $\gamma|P|$ and therefore alter the leading coefficient. It would require a relative shift of 0.47750 to reconcile these particular targets, and is not a vacuum-energy constant or momentum origin. The frame package explicitly keeps the inherited classical time scale; no such rescaling is made here. **(d)** The good det package does not compute the renormalized continuum vacuum sound speed or its bad quantum matching. A comparison after matching both sound speeds is therefore not completed by subtracting Toda's sound shift alone.

6 What this establishes, checks, and failures recorded

Verdict. (c) This is evidence against identifying either the formally continued trace or the explicitly evaluated stable-band trace with the Toda one-body dispersion in the inherited quantum units and tested asymptotic fit class. It is a one-body mismatch and cannot be blamed on the undetermined antisymmetric two-body phase frame. It also shows that the proposed unchanged $-2/\pi$ target is not a valid bad-equation target.

(d) It is *not* a refutation of the good-equation conditional charge invariant, nor an unconditional refutation of bad-Boussinesq quantization. The full bad trace requires an extra prescription; the projected determinant and quantum charge/background matching remain open; and finite-y extrapolations do not prove the required iterated limit. The physical comparison is therefore **inconclusive**, with a large, explicit conditional numerical discrepancy. Neither success nor failure of this one-body test could select p_{mode} over $p_{\text{can}} = \partial_v L_1$: their common invariant is exactly the tested quantity. A successful test would support only that common invariant.

Claim	Label	Check/status
Action, Planck and good/bad dictionaries	(c)	`onebody_continuum.py`; symbolic scaling
Formal bad F_{ac} and $+2/\pi$	(c)/(d)	`onebody_continuum.py`; continuation assumption
Stable-band trace and tadpole sign	(c)	`onebody_continuum.py`; exact primitive and quadrature
Complete physical bad determinant	(d)	unstable band/domain unresolved
Toda fixed-dressed-P coefficients and velocity	(c)	`onebody_toda.py`; stored data reanalysis
Large-y coefficient 0.15912 ± 0.00334	(c)	numerical, model-dependent envelope
Origin/mass-counterterm invariant	(c)	`onebody_continuum.py`; exact annihilator
Physical charge identification and limit exchange	(d)	not established

Failures recorded.

1. A scratch derivation initially inverted the Hirota transmission ratio, giving the wrong sign of the stable-band trace. It was corrected to $-\mathcal{H}(\text{difference})/\mathcal{H}(\text{sum})$ before the final checker. The corrected expression was then checked against the potential fluctuation equation and a symbolic antiderivative; no tolerance was relaxed.

2. The inherited bad-equation note's positive normal-ordering energy fails the bad Hamiltonian sign check. The corrected sign is negative; the difference is explicitly $6k/\pi$, not hidden in an energy constant.

3. A full real-frequency bad trace using only the good det assumptions cannot be constructed: $|q| > 1$ has unstable frequencies. The extra continuation/reduction assumption is retained as an obstruction, not replaced by an unannounced regulator.

4. The hoped-for equal small-amplitude coefficients fail for the specified conditional targets. Both the sign change under formal continuation and the stable-band resonance are retained.

5. The Toda fits do not establish a uniform semiclassical limit or an exact $1/(2\pi)$ coefficient. Their envelopes are not rigorous confidence intervals; the unmatched-limit systematic has no assigned numerical bound.

6. The historical scripts were not rerun: some write existing data, and this task authorizes new `onebody_*` files only. No existing result is newly certified. Both new checkers pass their stated tests; no final assertion tolerance was weakened.

7. The first LaTeX layout had two overlong text lines. Break opportunities and a shorter paragraph heading fixed them before the final rendering.

8. An added symbolic endpoint check initially failed structural equality: SymPy's factorization returned an unevaluated $0\sqrt{3}$. Exact simplification reduced it to zero. Neither the identity nor a tolerance was changed.

Reproduction and artifacts. Run from `phase2/scripts/`:

```
PYTHONDONTWRITEBYTECODE=1 python3 onebody_continuum.py
PYTHONDONTWRITEBYTECODE=1 python3 onebody_toda.py
```

Each writes only its own `onebody_*_results.json` and ends with `ALL CHECKS PASSED`. Runtime is seconds to tens of seconds on the task machine. Input hashes and fit details are saved in the Toda output. Build this document with `pdflatex` twice from `phase2/`. No existing file, README status line, or git state is changed.